INCH-POUND

MIL-C-49285/27
19 October 1989

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, ONE INDIVIDUALLY SHIELDED PAIR OF 24 (7 X 32) AWG

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.

Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	22 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	42 pF/ft
Continuous working voltage	300 V rms maximum between conductors
Shield dc resistance	20.5 ohms/1,000 feet maximum

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TABLE II. Description, physical.

Components	Construction details
Number of pairs	One individually shielded
Conductor type and wire size	Stranded tinned copper 24 AWG
Conductor stranding	7 X 32 AWG
Conductor insulation	Polyethylene
Conductor insulation thickness	0.016 inch nominal thickness
Drain wire type and size	Stranded tinned copper 24 AWG
Drain wire stranding	7 X 32 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.025 inch
Finished cable diameter	0.176 inch maximum outside diameter
Cable style (UL)	2092
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	8.0 twists per foot ±10 percent

REQUIREMENTS:

Design and construction: (See table II).

Shield color code. The shield color will be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located between the shield and the jacket. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Conductor color code. The first conductor is black; the second conductor is clear.

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Shield integrity test.*

Frequency	Response **
100 kHz	113 dB below reference
500 kHz	104 dB below reference
1 MHz	98 dB below reference
5 MHz	84 dB below reference
10 MHz	78 dB below reference
15 MHz	75 dB below reference
20 MHz	70 dB below reference
25 MHz	67 dB below reference
30 MHz	59 dB below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 46 dB below reference.

Flammability. The cable shall pass UL 1581 Vertical-Tray flame test requirements.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable shall have a durometer hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/27.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1114-28)

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